

بسم الله الرحمن الرحيم

King Abdulaziz University
Faculty of Computing and information Technology
Computer Science Department



CPCS-214 Computer Architecture and Organization

19 - 32-Bit Immediate Operands

Introduction

- Although constants are frequently short and fit into the 16-bit field, sometimes they are bigger

LUI Instruction

- MIPS includes instruction *load upper immediate* (*lui*)
- Sets upper 16 bits of a constant in a register
- Subsequent instruction specifies the lower 16 bits of constant



LUI Instruction

The machine language version of `lui $t0, 255` is register 8:

001111	00000	01000	0000 0000 1111 1111
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Contents of register `$t0` after executing `lui $t0, 255`:

0000 0000 1111 1111	0000 0000 0000 0000
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Example

- What is the MIPS assembly code to load 32-bit constant into register \$s0?

0000 0000 0011 1101 0000 1001 0000 0000

Example

- First, load upper 16 bits, which is 61 in decimal, using lui:

Upper: 61	Lower: 2304
<u>0000 0000 0011 1101</u>	<u>0000 1001 0000 0000</u>

lui \$s0, 61 #61_{ten} = 0000 0000 0011 1101_{two}

- Value of register \$s0 afterward is

0000 0000 0011 1101 0000 0000 0000 0000

Example

- Next is to insert lower 16 bits, whose decimal value is 2304:

`ori $s0, $s0, 2304` $\#2304_{\text{ten}} = 0000\ 1001\ 0000\ 0000_{\text{two}}$

- Final value in register \$s0 is the desired value:

0000 0000 0011 1101 0000 1001 0000 0000

Notices

- Creating 32-bit constants needs care
- Instruction `addi` copies the left-most bit of 16-bit immediate field of the instruction into the upper 16 bits of a word
- *Logical or immediate* loads 0s into upper 16 bits and hence is used by the assembler in conjunction with `lui` to create 32-bit constants